Assumptions of Major Machine Learning Models

Linear Regression

- Linear relationship between predictors and target variable
- Independence of errors
- Homoscedasticity (constant variance of residuals)
- Residuals approximately normally distributed
- No strong multicollinearity

Ridge Regression

- Linear relationship between features and target
- Features should be scaled
- Multicollinearity may exist but coefficients are regularized

Lasso Regression

- Linear relationship between predictors and target
- Features should be scaled
- Sparse representation may exist (some coefficients can become zero)

Elastic Net

- Linear relationship between features and target
- Features should be scaled
- Combination of L1 and L2 regularization assumptions

Logistic Regression

- Dependent variable is categorical (usually binary)
- Linear relationship between predictors and log-odds
- Independent observations
- No strong multicollinearity
- Large sample size preferred

Support Vector Machine (SVM)

- Data may be linearly separable in original or transformed space
- Features should be scaled

- Independent observations
- A meaningful margin exists between classes

K-Nearest Neighbors (KNN)

- Similar points exist close to each other in feature space
- Distance metric represents similarity properly
- Features should be scaled
- Local structure exists in the data

Naive Bayes

- Conditional independence of features given the class
- Features follow a probability distribution depending on the variant
- Each feature contributes independently to prediction

Decision Trees

- Data can be partitioned recursively
- Features contain enough information to split classes
- No assumption about underlying data distribution

Random Forest

- Ensemble of multiple decision trees reduces variance
- Bootstrap sampling introduces randomness
- Random feature selection improves generalization

Gradient Boosting

- Weak learners can be improved sequentially
- Errors from previous models can be minimized step by step
- Additive models approximate complex relationships

AdaBoost

- Weak learners perform slightly better than random guessing
- Misclassified samples should receive higher weights in next iteration

XGBoost

- Additive tree-based learners approximate complex relationships

- Regularization helps reduce overfitting
- Sequential error correction improves performance

LightGBM

- Gradient boosting framework with tree learners
- Leaf-wise tree growth can improve accuracy
- Large datasets improve performance

K-Means Clustering

- Clusters are spherical
- Clusters have similar variance
- Distance metric (usually Euclidean) represents similarity

DBSCAN

- Clusters are dense regions separated by sparse regions
- Distance metric properly measures similarity

Principal Component Analysis (PCA)

- Large variance directions capture most information
- Features should be scaled
- Linear relationships between variables

Linear Discriminant Analysis (LDA)

- Predictors follow multivariate normal distribution
- Equal covariance matrices across classes
- Observations are independent

Neural Networks

- Data contains learnable patterns
- Large datasets improve performance
- Features should be normalized or scaled